

$$d(x, y) = |x - y|,$$

The open sphere $S(a, r)$, with centre $a \in \mathbb{R}$, and $r > 0$, a real number, is given by

$$S(a, r) = \{x \in \mathbb{R} : d(x, a) = |x - a| < r\}, \text{ and}$$

$S^c(a, r)$, the closed sphere with centre $a \in \mathbb{R}^2$, and $r > 0$, a real number, is given by

$$S^c(a, r) = \{x \in \mathbb{R} : d(x, a) = |x - a| \leq r\}.$$

Example 15.4:

In the metric space, $(\mathbb{R}^2, d_E)$, where, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, and d_E , the Euclidean distance

$$d_E(x, y) = \sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2]}, \text{ for } x = (x_1, x_2), \text{ and } y = (y_1, y_2),$$

The open sphere $S(a, r)$, with centre $a = (a_1, a_2) \in \mathbb{R}^2$, and $r > 0$, a real number, is given by

$$S(a, r) = \{x \in \mathbb{R}^2 : d_E(x, a) = \sqrt{[(x_1 - a_1)^2 + (x_2 - a_2)^2]} < r\}, \text{ and}$$

$S^c(a, r)$, the closed sphere with centre $a \in \mathbb{R}^2$, and $r > 0$, a real number is given by

$$S^c(a, r) = \{x \in \mathbb{R}^2 : d_E(x, a) = \sqrt{[(x_1 - a_1)^2 + (x_2 - a_2)^2]} \leq r\}.$$

15.2.3.1.2 Open set, Closed set of arbitrary metric space

Definition Open set of a metric space (X, d) : A subset T of X is said to be an *open set*, if for each element x of T , there is an open sphere $S(x, \epsilon)$, with $\epsilon > 0$ such that $S(x, \epsilon) \subseteq T$. In other words, every member of T is *well inside* T , and hence, no member is a boundary point.

Examples and Non-Examples

- i) $]1, 3[\cup]5, 7[$ is an open set; because (a) every open interval is an open set, and (b) union of open sets is open (*Both (a) and (b) are proved in one of the theorems that follow*)
- ii) The semi-closed interval $X = [1, 3[$ is NOT an open set, as no open sphere $]1 - \epsilon, 1 + \epsilon[$, with $\epsilon > 0$, can be contained in $[1, 3[$. Similarly, closed interval $X = [1, 3]$ is NOT an open set, as the no open sphere $]1 - \epsilon, 1 + \epsilon[$, with $\epsilon > 0$, can be contained in $[1, 3]$.
- iii) For a given real number a each of

(iii-b) $] - \infty, a[$ and (iii-c) $]a, \infty[$ is an open set:

For (iii-b): Let $x \in] - \infty, a[$, then $x < a$. Let $\epsilon = a - x > 0$.

Then $S(x, \epsilon) \subseteq] - \infty, a[$. Hence, the proof of the claim.

For (iii-c):

Let $x \in]a, \infty[$, then $a < x$. Let $\varepsilon = x - a > 0$. Then $S(x, \varepsilon) \subseteq]a, \infty[$. Hence, the proof of the claim.

Next, we establish some significant properties of open sets.

Theorem 15.2.3.1.2.1: In a metric space (X, d)

- i) **Union of arbitrary number of open sets is an open set.**
- ii) **intersection of finite number of open sets is open**
- iii) **The universal set X is an open set**
- iv) **$\emptyset$, the empty set, is open,**
- v) **Every open sphere $S(a, r)$ is an open set,**

Note: Here, we prove (i) and (ii) only; (iii) and (iv) are proved under check your progress; and

v) is proved under Exercises and Answers

Proof

- i) **Union of arbitrary number of open sets is an open set**

Let each X_i , be open for $i \in I$, where I is an index set, which may be finite or infinite

To show $\bigcup_{i \in I} X_i$ is open.

Let $x \in \bigcup_{i \in I} X_i$, then $x \in X_j$, for some particular j . As X_j is open, therefore there is an $\varepsilon > 0$

such that $S(x, \varepsilon) \subseteq X_j$ and $X_j \subseteq \bigcup_{i \in I} X_i$. Therefore, for every $x \in \bigcup_{i \in I} X_i$, Hence, $S(x, \varepsilon) \subseteq \bigcup_{i \in I} X_i$.

- ii) **Finite intersection $\bigcap_{i=1}^n X_i$ of open sets is an open set**

Let $x \in \bigcap_{i=1}^n X_i$, where each X_i is open

If $\bigcap_{i=1}^n X_i$ is $\emptyset$, the empty set, then, by (iv) it is open.

else, if $\bigcap_{i=1}^n X_i$ is NOT $\emptyset$, let $x \in \bigcap_{i=1}^n X_i$,

$x \in X_i$, for each X_i , which is an open set, for $i = 1, 2, \dots, n$

As each X_i , is open, for $i = 1, 2, \dots, n$, there exists $\varepsilon_i > 0$ such that

$$S(x, \varepsilon_i) \subseteq X_i \text{ for } i = 1, 2, \dots, n.$$

Let $\varepsilon = \min \{\varepsilon_i\}$ for $i = 1, 2, \dots, n$.

Then, $S(x, \varepsilon) \subseteq S(x, \varepsilon_i) \subseteq X_i$ for each $i = 1, 2, \dots, n$

$$S(x, \varepsilon) \subseteq \bigcap_{i=1}^n X_i,$$

Thus, for every $x \in \bigcap_{i=1}^n X_i$, $S(x, \varepsilon) \subseteq \bigcap_{i=1}^n X_i$, **Hence $\bigcap_{i=1}^n X_i$ is open.**

Definition Closed set of a metric space (X, d) : A set $T \subseteq X$ is said to be a closed if its complement $X - T$ is an open set in X .

Examples 15.5:

- i) $[1, 3] \cup [5, 7]$ is a closed set, because **(a)** every closed interval is a closed set, and **(b)** union of closed sets is closed
(Both (a) and (b) are proved in one of the theorems that follow)
- ii) The semi-closed interval $X = [1, 3[$ is NOT a closed set, because its complement $] - \infty, 1[\cup [3, \infty[$ is not open as no sphere $S(3, \varepsilon) =] 3 - \varepsilon, 3 + \varepsilon[$ can be contained in $] - \infty, 1[\cup [3, \infty[$.

- iii) For a given real number a each of

(iii-b) $] - \infty, a]$ and **(iii-c)** $[a, \infty[$ is closed:

For (iii-b): For $] - \infty, a]$, its complement is $]a, \infty[$, which is open.

Hence, the given set is closed.

For (iii-c):

For $[a, \infty[$, its complement $] - \infty, a[$ is open.

Hence, the proof of the claim.

Theorem 15.2.3.1.2.2: In a metric space (X, d) ,

- i) **union of a finite number of closed sets is a closed set.**
 ii) **arbitrary intersection of closed sets is open**
 iii) **the universal set X is a closed set**
 iv) **$\emptyset$, the empty set, is a closed set,**

(From Theorems 15.2.3.1.2.1 and 15.2.3.1.2.2, we may observe that the Universal set X , and the empty set $\emptyset$, each is both an open set and a closed set.)

Here, we prove (i) and (ii) only; (iii) and (iv) are proved under check your progress.

Proof:

- i) **Union of a finite number of closed sets is a closed set.**

Let $C_i, i = 1, 2, \dots, n$ be n closed sets.

Consider the complement of their union

$$X - \left(\bigcup_{i=1}^n C_i \right), \quad \text{which, by De Morgan's Law,} \\ = \bigcap_{i=1}^n (X - C_i), \quad \text{where each } X - C_i, \text{ by definition, is open.}$$

Also, we have proved a finite intersection of open sets is open.

Hence, the proof.

- ii) **intersection of close sets is closed**

The proof is exactly on the lines of the above proof.

Let $C_i, i \in I$, be arbitrary number of closed sets.

Consider the complement

$$X - \bigcap_{i \in I} C_i \quad \text{by De Morgan's Law,}$$

$= \bigcup_{i \in I} (X - C_i)$, where each $X - C_i$, by definition, is open.

Also, we have proved union of open sets is open.

Hence, the proof.

Check Your Progress 2

- 1) For the metric space (X, d) , with

$d: X \times X \rightarrow \mathbb{R}$, such that for $x, y \in X$,

$$d(x, y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y, \end{cases}$$

find $S(a, 1)$ and $S^c(a, 1)$, respectively the open and closed spheres of radius 1, and centred at a .

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- 2) Prove that the universal set X is an open set

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- 3) Prove that $\emptyset$, the empty set, is open,

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- 4) Prove that the universal set X is a closed set

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- 5) Prove that $\emptyset$, the empty set, is a closed set.

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15.2.4 Convergence, continuity, connectedness and compactness in metric space

In Unit 9 of the previous block, and Unit 12 of the current block, we discussed the concepts of *convergence* of sequences, and *continuity of functions*, for $\mathbb{R}$, set of real numbers, equipped with Euclidean metric, and then extended these concepts to $\mathbb{R}^n$, the n -dimensional Euclidean space.

Here, we extend these concepts to **metric space**, and in the next section, extend these to **topological space**.

Our discussion is limited to just introduction of the concepts, and to the statements of some relevant and directly applicable theorems without proofs.

15.2.4.1 Convergence of a sequence in metric space

Definition Let (X, d) be a metric space, with $X \neq \emptyset$, and metric d on X . Further, let $(x_n) = (x_1, x_2, \dots, x_n, \dots)$ be an infinite sequence of points of X . Then, we say that **the sequence (x_n) is convergent** if there exists a point x in X such that for each $\varepsilon > 0$, there exists a positive integer k , such that $d(x_n, x) < \varepsilon$; for all $n > k$.

In the case of a metric space, the definition given above can be equivalently stated (*without proof*) as follows:

Theorem 15.2.4.1.1: In a metric space, a sequence $(x_n) = (x_1, x_2, \dots, x_n, \dots)$ of points of X . **is said to be convergent** if for each open sphere $S(x, \varepsilon)$, $\varepsilon > 0$, centered on x , there exists a positive integer k such that x_n is in $S(x, \varepsilon)$ for all $n > k$.

Next, we extend the concept of **continuity** of a real-valued real function to that of a function from one **metric space** to another metric space.

15.2.4.2 Continuity of a function in metric space

Let (X_i, d_i) , for $i = 1, 2$, be two metric spaces. Then a function $f: X_1 \rightarrow X_2$

is said to be **continuous** at point $x = x_0$ of X_1 ,

if for each $\varepsilon > 0$, **there exists** a real number $\delta > 0$, such that for all points x in X_1 , satisfying

$$d_1(x, x_0) < \delta \text{ imply } d_2(f(x), f(x_0)) < \varepsilon.$$

It needs to be noted that points $f(x)$ and $f(x_0)$ belong to X_2 .

The above definition of continuity is exactly on the same lines as that of continuity of real valued real function discussed in Unit 9. The examples and problems discussed over there can be included here also, because this definition of continuity subsumes the definition of continuity given in Unit 9.

Next, we state two theorems, without proof, about continuity of a function between **metric spaces**.

Theorem 15.2.4.2.1: Let (X_i, d_i) , for $i = 1, 2$, be two metric spaces. Then a function

$$f: X_1 \rightarrow X_2$$

is **continuous** at point $x = x_0$ of X_1 , if and only if

for each **open sphere**, say S_2 in X_2 , with centre $f(x_0)$ and radius $\epsilon > 0$, there is an **open sphere** S_1 in X_1 having centre x_0 and radius $\delta > 0$ such that $f(S_1)$, the image of S_1 , in X_2 , is contained in the sphere S_2 .

As every open sphere is an open set, the above theorem may be stated as

Theorem 15.2.4.2.2: Let (X_i, d_i) , for $i = 1, 2$, be two metric spaces. Then a function

$$f: X_1 \rightarrow X_2$$

is **continuous** at point $x = x_0$ of X_1 , if and only if

for each non-empty **open set**, say S_2 in X_2 , there is an **open set** S_1 in X_1 such that $f(S_1)$, the image of S_1 , in X_2 , is contained in S_2 .

Theorem 15.2.4.2.3: Let (X_i, d_i) , for $i = 1, 2$, be two metric spaces. Then a function

$$f: X_1 \rightarrow X_2$$

is **continuous** at point $x = x_0$ of X_1 , **if and only if** for every sequence $(x_n) = (x_1, x_2, \dots, \dots)$ which converges to x_0 , then the sequence $(f(x_n))$ converges to $f(x_0)$ in X_2 , i. e., symbolically,

$$\text{if and only if} \quad x_n \rightarrow x_0 \text{ implies } f(x_n) \rightarrow f(x_0).$$

15.2.4.3 Connectedness of a set in a metric Space

A **set** S in a metric space (X, d) is said to be **connected**, if it cannot be represented as the union of two or more **disjoint** non-empty open sets in the metric space.

We state some theorems without proof.

Theorem 15.2.4.3.1 A set S in a metric space (X, d) is **connected**, if and only if it cannot be represented as the union of two or more disjoint non-empty **closed** sets.

Theorem 15.2.4.3.2 A subset of R , equipped with Euclidean metric, is **connected** if and only if it is an **interval**—whether open, semi-open, or closed.

Example 15.6:

Example 1: In the Euclidean (metric) space R of real numbers, each of the intervals $]3, 7[$, $[3, 7[$, $[3, 7]$ is a connected set. **In general, any interval—whether open, semi-open, or closed—is connected.**

Example 15.7: The set $S =]3, 7[\cup]10, 16[$ is **NOT** connected in Euclidean (metric) space R of real numbers. **In general, union of two non-overlapping intervals—whether open, semi-open, or closed—is not connected.**

15.2.4.4 Compactness of a set in a metric space

In the previous unit, we have already discussed the concept of *compactness* for Euclidean space $\mathbb{R}$, where we mentioned that **a subset of $\mathbb{R}$ is compact, if it is closed and bounded**. There, we also discussed the related Bolzano-Weierstrass Theorem and its Economic Applications. Here, we briefly discuss extension of the concept of compactness to a **general metric space**, and in the next section, we extend the concept further to a **topological space**.

We already know that, in a metric space, a **closed set** is complement of an open set. In order to know **what is compact set** in a **metric space**, let us know what is **bounded set** in a metric space, through the

Definition: A subset S of X in a metric space (X, d) is called **bounded** if there exists $r > 0$ such that for all s and t in S , we have $d(s, t) < r$. The metric space (X, d) is a bounded metric space (or d is a bounded metric) if X is bounded as a subset of itself.

Example 15.8: Every **finite** set in a metric space is compact, i. e. it is closed and bounded.

Let $\{a_1, a_2, \dots, a_n\}$ be a finite set of metric space (X, d) . **Further**, let

$$M = \{d(x_i, x_j), i, j = 1, 2, \dots, n, \text{ then,}$$

by definition, for any pair (x_i, x_j) , we have $d(x_i, x_j) \leq M$

Thus, a finite set is a bounded set.

Next, we show $X - \{a_1, a_2, \dots, a_n\}$ is a closed set.

For this purpose, first of all, we know that the universal set X of metric space (X, d) is open. Consider

$$X - \{a_1, a_2, \dots, a_n\} = (X - \{a_1\}) \cup (X - \{a_2\}) \cup \dots \cup (X - \{a_i\}) \cup \dots \cup X - \{a_n\},$$

The set $\{a_1, a_2, \dots, a_n\}$ would be closed, if we show each $X - \{a_i\}$ is open set, as follows:

Let $y \in X - \{a_i\}$, then $y \neq a_i$. Let $d(y, a_i) = \varepsilon$, and then $\varepsilon > 0$.

Then $S(y, \varepsilon/2)$ does not contain a_i . In other words,

The sphere $S(y, \varepsilon/2)$ is completely contained in $X - \{a_i\}$. Hence,

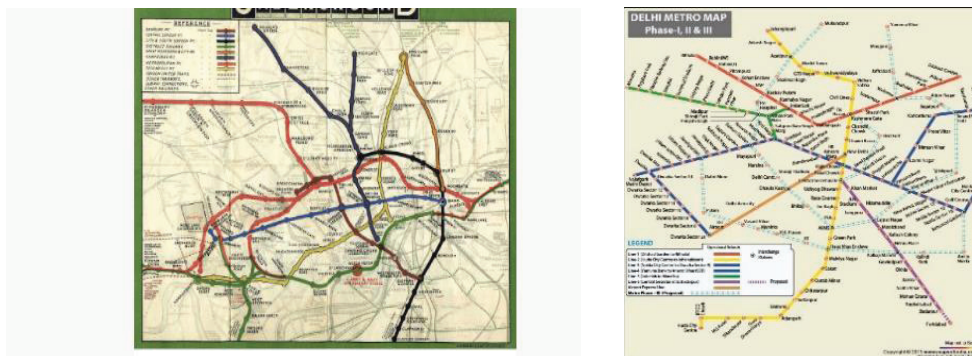
$$X - \{a_i\}, \text{ is open for } i=1, 2, \dots, n,$$

And therefore, the set $\{a_1, a_2, \dots, a_n\}$ is closed.

Next, we discuss the topic of **Topology**, which in one sense is further generalization of the concept/discipline of **Metric Space**.

15.3 POINT SET TOPOLOGY

Let us again look at the two figures below, where the LHS gives a sort of aerial view of Delhi Metro Network, and the RHS shows a type of map available within a Metro rail.



As mentioned earlier, by losing sense of exact distance and exact angle, we get a map which is much more comprehensible to the travellers.

Also, in real life problems, we encounter situations, for which no useful metrics are defined, or even can be defined, but which—for solution of the relevant problems—**require the concept of continuous function to be used**. For such situations/ problems, it is necessary for us to liberate our concept of continuity from its dependence on metric spaces.

The **discipline of topology**, being a generalization of both metric space and of Euclidean geometry, **ignores**

- i) the *concept of distance*, between two points, which forms the basis of *Metric space*, and
- ii) the concepts of *distance*, *angle*, and *geometric congruence* etc. of *Euclidean Geometry*.

Topology uses the concept of open set as its fundamental concept. The concept of ‘open set’ was discussed for a *general metric space*, and also for various *Euclidean geometries*. In all of these, the concept of distance was used in defining *open sphere*, which, in turn was used to define the concept of *open set*.

Repeating for the sake of emphasis, in topology, we take the concept of OPEN SET as its fundamental concept, completely ignoring the concepts of distance/metric, angle, geometric congruence etc.

Most of the concepts and theorems etc. of Metric space—based upon, or using—the concept of OPEN SET, may be generalized, or carried over almost directly, to the corresponding ones in Topology. Also, in this respect, Theorem 15.2.4.2.2 shows that crucial concept of ‘continuity’ can be also discussed in terms of open sets only.

15.3.1 Basic Definitions

The concept/ discipline of topology on a set X is defined, developed and discussed in terms (i) subsets of X , and (ii) unions and intersection operations on the subsets.

For defining the concept of topology, briefly we recall, from Unit1, the concept of Powerset of a set:

The set of all subsets of a given set A is called **Power Set of A** , and is denoted as $P(A)$.

For example, if $A = \{1, 2, 3\}$, then $P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$.

Topology: For a given non-empty set X , *a topology on X is the ordered pair (X, T)* , where T is a collection

of subsets of X , (i. e. $T \subseteq P(X)$, the power set of X), satisfying the following conditions

- i) the universal set $X \in T$
- ii) the empty set $\emptyset \in T$
- iii) the union, of any number of sets from T , is a set in T ;
- iv) the intersection, of any *finite* number of sets from T , is a set in T .

In other words, (X, T) is topology, if T is closed under (i) union of sets, and (ii) finite intersection of sets, apart from T having X and $\emptyset$ as its members.

A topology (X, T) is also called a Topological Space.

Open Set: Each set in T is called an *open set* of the topological space (X, T) . And, a set is open only if it belongs to T . Thus, *in every topology, the universal set X , and the empty set $\emptyset$ are open sets.*

Closed Set: A subset S of X is said to be closed, if its complement $X - S$ is in T . Thus, *in every topology, the universal set X , and the empty set $\emptyset$ are closed sets also.*

Neighbourhood of a point x , or a set S , in topological space (X, T) : A neighborhood of a point x (or a set S), in a topological space is an open set (i. e., is a member of T) which contains the point x (or the set S).

It may be noted that that for different collections of subsets of X , we get different topologies on X .

Example 15.9: Let X be a non-empty set, and let $T = \{\emptyset, X\}$, where $\emptyset$ is the empty set, and X is the universal set. The (X, T) is a topology.

The first two conditions (i) the universal set $X \in T$, and (ii) the empty set $\emptyset \in T$, are obviously satisfied.

The other two topological conditions are satisfied, as explained below.

iii) Only possible union of sets in T is $X \cup \emptyset = X$, which belongs to T

iv) Only possible intersection of sets in T is $X \cap \emptyset = \emptyset$ which belongs to T .

Example 15.10: The ordered pair $(X, P(X))$, is a topology on X , where X is a non-empty set, and $P(X)$ is the power set of X .

The first two conditions (i) the universal set $X \in T$, and (ii) the empty set $\emptyset \in T$, both being in $P(X)$ are satisfied.

The other two topological conditions are satisfied, as explained below:

As $P(X)$ is the set of all subsets of X , hence,

iii) union of any number of subsets of X —being itself a subset of X —is in $P(X)$.

iv) intersection of any number of subsets of X —being itself a subset of X —is in $P(X)$.

This topology is called the *discrete topology* on X .

Example 15.11: Let (X, d) be a metric space, and $T = \{S: S \text{ is an open set in the metric space } (X, d)\}$. Then, (X, T) is a topological space. **The topological space (X, T) is called topology induced by the metric space (X, d) , or topology on X induced by the metric d .**

In a metric space (X, d) , the universal set X , and the empty set $\emptyset$ are always open, hence both belong to T .

The other conditions are satisfied by *Theorem 15.2.3.1.2.1*, which states that in a metric space (X, d)

i) *Union of arbitrary number of open sets is an open set.*

ii) *Intersection of finite number of open sets is open.*

15.3.2 Continuity, connectedness and compactness in topological space

Each of the above-mentioned concepts for topological space, is obtained by generalizing the corresponding concept for metric space.

15.3.2.1 Continuity of a function in topological space

Let (X_i, T_i) , $i = 1, 2$ be two topologies, and $f: X_1 \rightarrow X_2$ be a mapping of one topological space into the other. Then, f is said to be *continuous* at a point x_0 of X_1 , if for each neighbourhood H of $f(x_0)$ in X_2 , there exists a neighbourhood G of x_0 in X_1 such that $f(G) \subseteq H$.

15.3.2.2 Connectedness of a set in a topological space

The intuitive idea behind connectedness—is similar to that of continuity of *functions from* one domain to another; but here, *it is about a set in the domain*,